3. NETWORK SECURITY ESSENTIALS

DATE: 11/02/2026

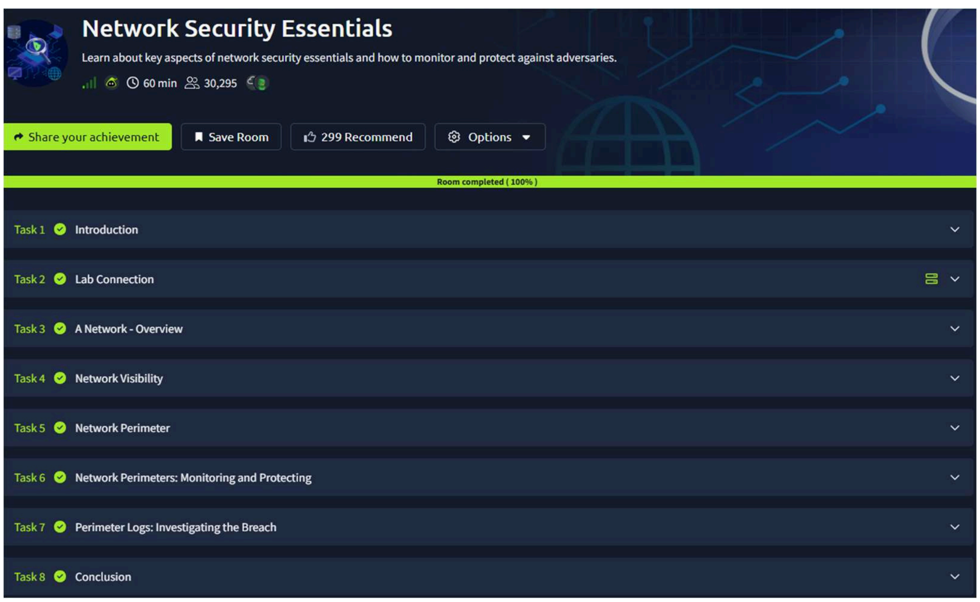
Aim:

To learn about key aspects of network security essentials and how to monitor and protect against adversaries.

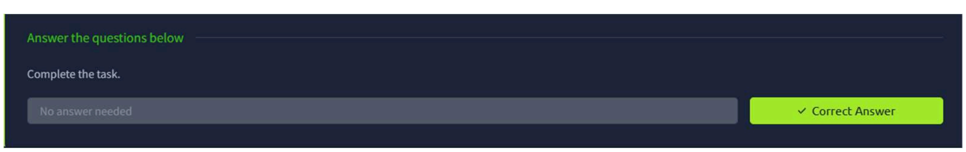
Procedure:

1. Log into TryHackMe and open the Network Security Essentials room.
2. Study basic network security concepts including firewalls, IDS, and IPS.
3. Understand common network threats such as malware, phishing, and DDoS attacks.
4. Learn monitoring techniques using logs and traffic analysis.
5. Explore methods to protect networks from adversaries.
6. Complete the tasks and answer the questions in the room.
7. Review the summary and confirm completion.

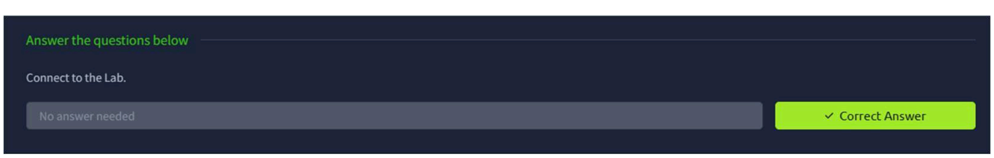
Tasks:



TASK 1 - Introduction



TASK 2 – Lab Connection



TASK 3 – A Network-Overview

Answer the questions below

Continue to the Next task.

No answer needed — ✓ Correct Answer

TASK 4 – Network Visibility

Answer the questions below

Continue to the next task.

No answer needed — ✓ Correct Answer

TASK 5 – Network Perimeter

Answer the questions below

Continue to the next task.

No answer needed — ✓ Correct Answer

TASK 6 – Network Perimeters : Monitoring and Protecting

Answer the questions below

Examine the firewall logs. Which IP address is performing the port scan?

203.0.113.10 — ✓ Correct Answer

In the WAF Logs, which single source IP is responsible for all the blocked web attacks?

198.51.100.12 — ✓ Correct Answer

In the VPN logs, how many brute-force attempts failed?

90 — ✓ Correct Answer

Which suspicious IP address was found attempting the brute-force attack against the VPN gateway?

45.137.22.13 — ✓ Correct Answer

firewall_logs.txt

```
2025-09-15 08:30:01 ALLOW TCP 192.0.2.115:51234 -> 10.0.0.50:443
2025-09-15 08:30:02 ALLOW TCP 198.51.100.88:60123 -> 10.0.0.51:80
2025-09-15 08:30:03 ALLOW TCP 192.0.2.120:54321 -> 10.0.0.50:443
2025-09-15 08:30:04 ALLOW TCP 198.51.100.45:49876 -> 10.0.0.51:80
2025-09-15 08:30:05 BLOCK TCP 203.0.113.10:50001 -> 10.0.0.20:21
2025-09-15 08:30:06 BLOCK TCP 203.0.113.10:50002 -> 10.0.0.20:22
2025-09-15 08:30:07 ALLOW TCP 192.0.2.115:51235 -> 10.0.0.50:443
2025-09-15 08:30:08 BLOCK TCP 203.0.113.10:50003 -> 10.0.0.20:23
2025-09-15 08:30:09 BLOCK TCP 203.0.113.10:50004 -> 10.0.0.20:25
2025-09-15 08:30:10 ALLOW TCP 198.51.100.92:51111 -> 10.0.0.50:443
2025-09-15 08:30:11 BLOCK TCP 203.0.113.10:50005 -> 10.0.0.20:53
2025-09-15 08:30:12 ALLOW TCP 192.0.2.130:60321 -> 10.0.0.51:80
2025-09-15 08:30:13 BLOCK TCP 203.0.113.10:50006 -> 10.0.0.20:80
```

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/task6$ grep -iE 'block|blocked' waf_logs.txt
timestamp=2025-09-22T09:14:10Z src_ip=198.51.100.12 action=BLOCK request="GET /products.php?id=12%20UN
ION%20SELECT%20user,pass%20FROM%20users" rule_id=942100 attack_type="SQL Injection"
timestamp=2025-09-22T09:14:46Z src_ip=198.51.100.12 action=BLOCK request="GET /search.php?q=<script>al
ert('XSS')</script>" rule_id=941100 attack_type="XSS"
timestamp=2025-09-22T09:15:42Z src_ip=198.51.100.12 action=BLOCK request="GET /../../../../etc/passwd"
 rule_id=930120 attack_type="Directory Traversal"
```

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/task6$ grep 'FAILED_AUTH' vpn_logs.txt | wc -l
90
```

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/task6$ grep 'FAILED_AUTH' vpn_logs.txt
 | head -n 5
2025-09-22 10:12:02 FAILED_AUTH TCP 45.137.22.13:31245 -> 10.0.0.1:443 (user 'ad
min')
2025-09-22 10:12:03 FAILED_AUTH TCP 45.137.22.13:31246 -> 10.0.0.1:443 (user 'ad
```

TASK 7 – Perimeter Logs : Investigating The Breach

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ grep "BLOCK" firewall.log | awk '{print $5}' | cu
t -d: -f1 | sort | uniq -c | sort -nr
    279 203.0.113.45
     46 203.0.113.10
     26 10.8.0.23
```

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/task6$ grep "BLOCK" firewall_logs.txt | awk '{print $7}' | c
ut -d: -f1 | sort | uniq -c | sort -nr
     16 10.0.0.20
```

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ awk '{print $4}' vpn_auth.log | sort | uniq -c |
sort -nr
    137 svc_backup
     30 jsmith
     29 bob
     24 alice
```

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ grep "svc_backup" vpn_auth.log
2025-09-03 02:19:40 203.0.113.45 svc_backup SUCCESS assigned_ip=10.8.0.23
2025-09-03 02:19:50 203.0.113.45 svc_backup SUCCESS assigned_ip=10.8.0.23
2025-09-04 16:45:24 203.0.113.45 svc_backup SUCCESS assigned_ip=10.8.0.181
2025-09-06 02:04:13 192.0.2.115 svc_backup SUCCESS assigned_ip=10.8.0.172
2025-09-06 09:23:51 203.0.113.10 svc_backup SUCCESS assigned_ip=10.8.0.135
2025-09-06 13:51:58 198.51.100.45 svc_backup SUCCESS assigned_ip=10.8.0.32
```

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ grep -i "smb" ids_alerts.log
2025-09-05 06:10:00 [**] [1:2000201:1] ET EXPLOIT Possible MS-SMB Lateral Movement [**] [Classificatio
n: Attempted Unauthorized Access] [Priority: 1] {TCP} 10.8.0.23:2001 -> 10.0.0.51:445
2025-09-05 07:00:00 [**] [1:2000206:1] ET EXPLOIT Possible MS-SMB Lateral Movement [**] [Classificatio
n: Attempted Unauthorized Access] [Priority: 1] {TCP} 10.8.0.23:2006 -> 10.0.0.20:445
2025-09-05 07:40:00 [**] [1:2000210:1] ET EXPLOIT Possible MS-SMB Lateral Movement [**] [Classificatio
```

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ grep -i "c2" ids_alerts.log
2025-09-11 01:00:00 [**] [1:2001000:1] ET TROJAN Possible C2 Beaconing [**] [Classification: A network
 Trojan was detected] [Priority: 1] {TCP} 10.0.0.60:30000 -> 198.51.100.77:4444
2025-09-11 07:00:00 [**] [1:2001001:1] ET TROJAN Possible C2 Beaconing [**] [Classification: A network
 Trojan was detected] [Priority: 1] {TCP} 10.0.0.60:30001 -> 198.51.100.77:4444
2025-09-11 13:00:00 [**] [1:2001002:1] ET TROJAN Possible C2 Beaconing [**] [Classification: A network
```

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ grep -i "exfil" ids_alerts.log
2025-09-18 23:00:00 [**] [1:2002000:1] ET INFO Possible HTTP POST Large Upload [**] [Classification: P
otential Data Exfiltration] [Priority: 2] {TCP} 10.0.0.51:40000 -> 198.51.100.77:8080
2025-09-19 03:00:00 [**] [1:2002001:1] ET INFO Possible HTTP POST Large Upload [**] [Classification: P
otential Data Exfiltration] [Priority: 2] {TCP} 10.0.0.51:40001 -> 198.51.100.77:8080
2025-09-19 07:00:00 [**] [1:2002002:1] ET INFO Possible HTTP POST Large Upload [**] [Classification: P
otential Data Exfiltration] [Priority: 2] {TCP} 10.0.0.51:40002 -> 198.51.100.77:8080
2025-09-19 11:00:00 [**] [1:2002003:1] ET INFO Possible HTTP POST Large Upload [**] [Classification: P
```

TASK 8 - Conclusion

Result:

Thus the TryHackMe room “Network SecurityEssentials” was completed successfully.